

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Second Semester of M.Sc. (Mathematics) Examination May 2018
MA725 Real Analysis

Date: 11/05/2018, Friday

Time: 10:00 a.m. to 10:45 a.m. Maximum Marks: 20

MCQ



Important Instructions:

1. Tick the correct answer and it should be written in question paper itself.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Each question carries one mark.
4. All notations are standard.

Q - I Choose the most appropriate answer from the given options for the following questions.

1. The condition for Lebesgue measurability of a subset $E \subseteq \mathbb{R}$ is _____, where m^* is the outer measure.
(a) $m^*(E) = m^*(E \cap G) + m^*(E \cap G^c), \forall E \subset \mathbb{R}$
(b) $m^*(G) = m^*(G \cup E) + m^*(G \cap E^c), \forall G \subset \mathbb{R}$
(c) $m^*(E) = m^*(E \cup G) + m^*(E \cap G^c), \forall E \subset \mathbb{R}$
(d) $m^*(G) = m^*(G \cap E) + m^*(G \cap E^c), \forall G \subset \mathbb{R}$
2. $\bigcup_{n=1}^{\infty} \left[a + \frac{1}{n}, b - \frac{1}{n} \right]$ is _____.
(a) $[a, b]$ (b) F_{σ} set (c) G_{δ} set (d) None of these
3. Consider the following statements:
(I) The length of an interval I is the difference of end points of the interval.
(II) The Lebesgue outer measure of an interval is its length.
Then _____.

- (a) Only (I) is true. (c) Both (I) and (II) are true.
 (b) Only (II) is true. (d) Neither (I) nor (II) is true.
4. If f is non negative function which is integrable over a set E . Then given $\epsilon > 0$, there is a $\delta > 0$ such that for every set $A \subset E$ with $m(A) < \delta$, _____.
 (a) $\int_A f = \epsilon$ (c) $\int_E f < \epsilon$
 (b) $\int_A f < \epsilon$ (d) None of these
5. Consider the simple function $\varphi = \sum_{i=1}^k c_i \chi_{E_i}$ on $E \subseteq \mathbb{R}$ where $E_i \subseteq E$ are Lebesgue measurable sets and c_i 's are non-negative. Then _____, where m is the Lebesgue measure.
 (a) $\int_E \varphi dm = \prod_{i=1}^k \varphi(c_i) m(E_i)$ (c) $\int_E \varphi dm = \sum_{i=1}^k \varphi(c_i) m(E_i)$
 (b) $\int_E \varphi dm = \prod_{i=1}^k c_i m(E_i)$ (d) $\int_E \varphi dm = \sum_{i=1}^k c_i m(E_i)$
6. Let E be the set of irrationals in $[-1, 1]$. Then $\int_{[-1,1]} \chi_E$ is _____.
 (a) 1 (b) 0 (c) ∞ (d) 2
7. A function $f : E \rightarrow [-\infty, \infty]$ on a measurable set is integrable if _____.
 (a) $\int_E f^+$ exists
 (b) $\int_E f^-$ exists
 (c) $\int_E f^+$ and $\int_E f^-$ both exist
 (d) either $\int_E f^+$ or $\int_E f^-$ exists
8. If f is of bounded variation on $[a, b]$ and $c \in [a, b]$ then _____.
 (a) $V(f)(a, b) < V(f)(a, c) + V(f)(c, b)$
 (b) $V(f)(a, b) > V(f)(a, c) + V(f)(c, b)$
 (c) $V(f)(a, b) = V(f)(a, c) + V(f)(c, b)$
 (d) None of these
9. The set having positive measure contains _____ non-measurable sets
 (a) Exactly one (c) infinitely many
 (b) finitely many (d) None of these
10. If $\{u_n\}$ is a sequence of non-negative integrable functions on a set of finite measure and $f = \sum_{n=1}^{\infty} u_n$ pointwise a.e. then _____.
 (a) $\int f = \int u_n$ (c) $\int f < \sum_{n=1}^{\infty} \int u_n$
 (b) $\int f = \sum_{n=1}^{\infty} \int u_n$ (d) $\int f > \sum_{n=1}^{\infty} \int u_n$

11. Let $P \subset \mathbb{R}$ be a non-measurable set. Then _____.
 (a) $m(E) > 0$, for $E \subset P$ (c) $m(P) = \infty$
 (b) there is a non-Borel subset of P (d) $m(P) = 0$
12. The function $f(x) = \underline{\hspace{2cm}}$ is not of bounded variation over $[-2018, 2018]$.
 (a) x (b) $\log x$ (c) e^x (d) $|x|$
13. If $f^+(x) = \max \{f(x), 0\}$ and $f^-(x) = \max \{-f(x), 0\}$ then _____.
 (a) $f = f^+ + f^-$ (c) $|f| = f^+ - f^-$
 (b) $f = f^+ - f^-$ (d) $|f| = f^- - f^+$
14. A set $A \subseteq \mathbb{R}$ contains a non-measurable set if _____.
 (a) $m^*(A) = \frac{1}{2}$ (c) $m^*(A) = -\frac{1}{2}$
 (b) $m^*(A) = 0$ (d) None of these
15. $m^*(\mathbf{C}) = \underline{\hspace{2cm}}$, where $\mathbf{C}$ is a Cantor set.
 (a) $m^*([0, 1]) - 1$ (c) $m^*([0, 1])$
 (b) $m^*(\mathbb{R})$ (d) None of these
16. If $f'(x) = 0$ a.e. then _____.
 (a) f is a constant function
 (b) $m\{x : f'(x) \neq 0\} = 0$
 (c) f is absolutely continuous function
 (d) None of these
17. _____ is not a convex function.
 (a) $y = x^2$ (c) $\cos x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
 (b) $\sin x, x \in (\pi, 2\pi)$ (d) $\log x$
18. The Lebesgue outer measure m^* of a set $A \subseteq \mathbb{R}$ is _____.
 (a) $m^*(A) = \inf \{\sum (b_i - a_i) : \cup [a_i, b_i] \subseteq A\}$
 (b) $m^*(A) = \inf \{\sum (b_i - a_i) : \cup [a_i, b_i] \supseteq A\}$
 (c) $m^*(A) = \sup \{\sum (b_i - a_i) : \cup [a_i, b_i] \subseteq A\}$
 (d) $m^*(A) = \sup \{\sum (b_i - a_i) : \cup [a_i, b_i] \supseteq A\}$
19. If f is an absolutely continuous function on $[a, b]$, then _____.
 (a) f is of bounded variation (c) f is not continuous
 (b) f is not of bounded variation (d) None of these
20. The Lebesgue integral of a non-negative bounded measurable function over a countable set is _____.
 (a) 1 (b) 2 (c) ∞ (d) 0

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Date: 11/05/2018, Friday

Time: 10:45 a.m. to 01:00 p.m.

Maximum Marks: 50

Instructions:

1. **Section I and Section II must be written in separate answer books..**
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Figures to the right indicate marks.
4. All notations are standard.

SECTION I

Q - II (A) Attempt any TWO of the following. [8]

- (1) Let f be a function defined on a measurable set E . Then prove that f is measurable iff for any open set G in $\mathbb{R}$, the inverse image of $f^{-1}(G)$ is a measurable set.
- (2) Let f be a bounded function on $[a, b]$. If f is $\mathcal{R}$ -integrable, then prove that it is $\mathcal{L}$ -integrable. The converse of this statement is true? Justify.
- (3) "Monotone convergence theorem doesn't hold for decreasing sequence of functions." Is it true? Justify.

(B) Attempt any SIX of the following. [12]

- (1) State Littlewood's three principles.
- (2) Find $\overline{\lim}_{n \rightarrow \infty} (-n)$.
- (3) State Egoroff's theorem.
- (4) If f is measurable and $f = g$ a.e. then prove that g is measurable.
- (5) State Lusin's theorem.
- (6) If $\phi = \chi_{[-4,0]} - 2 \chi_{[-2,7]}$ then find $\int \phi$.
- (7) State Chordal slope lemma.
- (8) Prove that a convex function is absolutely continuous.
- (9) State Jensen's inequality.

SECTION II

Q - III (A) Attempt any THREE of the following.

[18]

- (1) Let $\{E_n\}$ be a sequence of measurable sets such that $E_1 \supset E_2 \supset \dots \supset E_n \supset \dots$. Let $m(E_1) < \infty$ then prove that $m\left(\bigcap_{i=1}^{\infty} E_i\right) = \lim_{n \rightarrow \infty} m(E_n)$. Justify if we drop the condition $m(E_1) < \infty$ from the statement.
- (2) Let E be a measurable set of finite measure and $\{f_n\}$ be a sequence of measurable functions defined on E . Let f be a real valued function such that for each $x \in E$, $f_n(x) \rightarrow f(x)$. Then prove that for given $\epsilon > 0$ and $\delta > 0$ there is a measurable set A in E with $m(A) < \delta$ and an integer N such that $\forall x \notin A$ and $\forall n \geq N$, $|f_n(x) - f(x)| < \epsilon$.
- (3) State and prove Fatous lemma. Does it hold for negative valued function? Justify your answer.
- (4) Let $\{f_n\}$ be a sequence of measurable functions convergent in measure to a function f . Then prove that there exists a subsequence $\{f_{n_k}\}$ of $\{f_n\}$ which converges to f a.e.
- (5) Prove that a function F is an indefinite integral over $[a, b]$ iff it is absolutely continuous over $[a, b]$.

(B) Attempt any TWO of the following.

[12]

- (1) Let f be a bounded real valued function defined over a set of finite measure. If $\inf\left\{\int_E \psi : \psi \text{ is a simple function and } f \leq \psi\right\} = \sup\left\{\int_E \phi : \phi \text{ is a simple function and } f \geq \phi\right\}$, then prove that f is measurable.
- (2) Prove that a function f is of bounded variation over $[a, b]$ iff it is a difference of two real valued monotone function on $[a, b]$.
- (3) Let f be an integrable function over $[a, b]$ and $F(x) = F(a) + \int_a^x f(t)dt$ then show that $F'(x) = f(x)$ a.e.